

Kubernetes App Lifecycle

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Agenda

- Overview of Deployment
- Deployment Strategies
- Lab: Managing Deployment
- Lab: Blue-Green Deployment Strategy



What is a Deployment in Kubernetes?

- A Kubernetes controller that manages **replica sets** and ensures desired state of pods.
- Automates:
 - Pod creation
 - Scaling
 - Rollouts/Rollbacks



Key Deployment Features

- Declarative updates
- Self-healing (replaces failed pods)
- Rollback capabilities
- Versioned updates



Benefits of Deployment

- Easy scaling
- Self-healing pods
- Rollback support
- Declarative and versioned



Rolling Update Strategy

- Default in Kubernetes
- Gradual replacement of old pods
- Zero downtime (if configured correctly)
- `maxUnavailable`, `maxSurge`

Default Values:

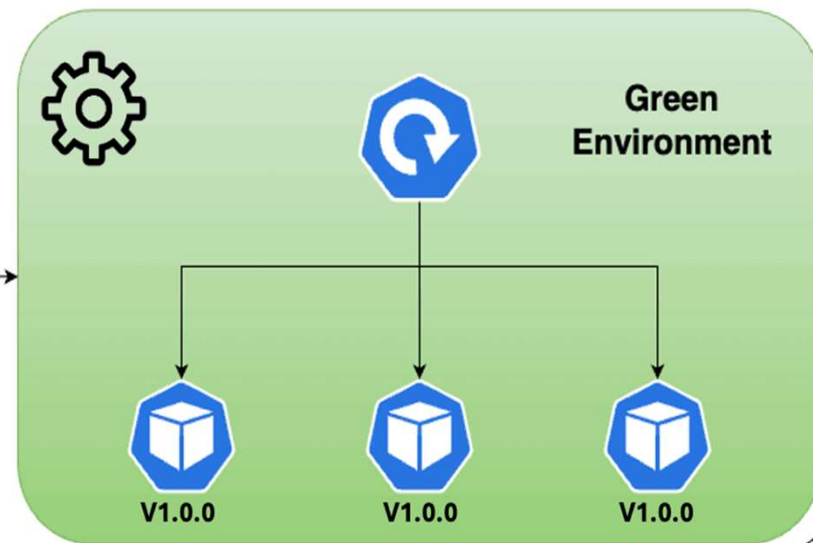
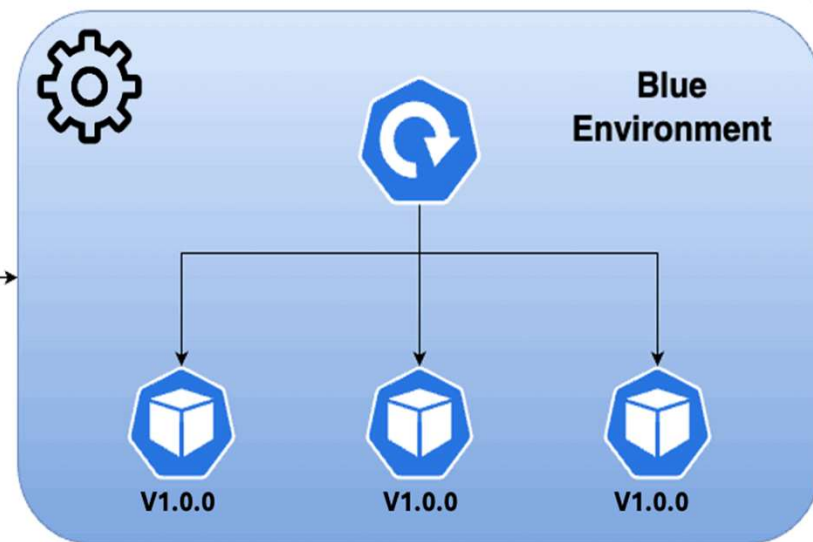
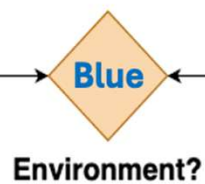
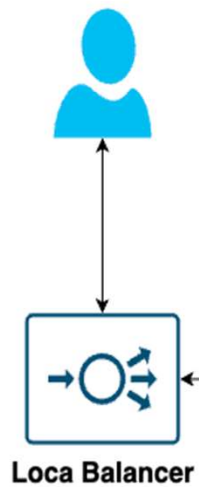
strategy:

type: RollingUpdate

rollingUpdate:

maxSurge: 25%

maxUnavailable: 25%



Blue-Green Deployment Strategy



Blue-Green Deployment

- Two environments: Blue (live), Green (new)
- Traffic switch via Service object
- Benefits: instant rollback, zero downtime
- Tools: **kubectl**, **Argo Rollouts**, custom scripts

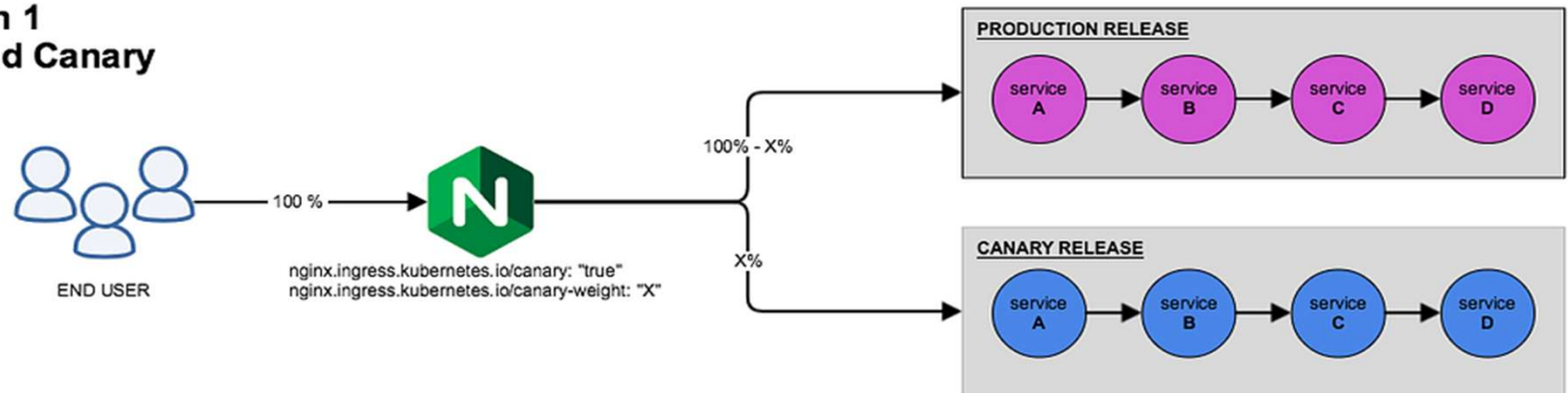


Recreate Strategy

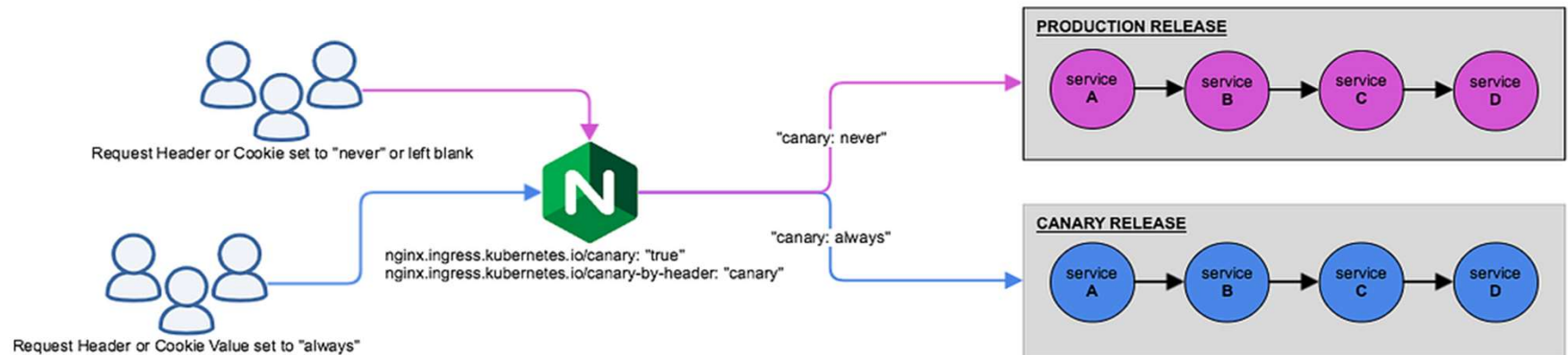
- All old pods terminated, then new ones created
- Downtime expected
- Use-case: incompatible app versions or DB migrations



Option 1 Weight-Based Canary



Option 2 User-Based Canary





Canary Deployment Strategy

- New version released to a **small subset of users first**
- Traffic gradually shifted from old → new version
- Used to **detect issues before full rollout**
- Requires traffic control (Ingress / Service Mesh)

Traffic Flow Example:

- v1 → 90% users
- v2 → 10% users → then 25% → 50% → 100%



Canary Deployment Strategy

Benefits

- Reduced risk
- Real production testing
- Safer than rolling update for critical systems

Tools

- NGINX Ingress Canary annotations
- Istio / Linkerd
- Argo Rollouts



There are more deployment strategy....



Strategy Comparison Slide

Strategy	Downtime	Risk Level	Traffic Control	Best For
Recreate	Yes	High	None	DB schema breaking changes
Rolling Update	No	Medium	Pod-level	Standard app updates
Blue-Green	No	Low	Service switch	Instant rollback
Canary	No	Very Low	% traffic split	Safe prod testing
A/B Testing	No	Low	User-based	Feature experiments
Shadow	No	Very Low	Mirrored traffic	Performance validation